

TASK-04

Create and use a Docker volume to share data between the host system and a container.

```
PS C:\Users\VICTUS> cd C:\docker-volume-demo
PS C:\docker-volume-demo> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c1a698e48bee	ubuntu	"bash"	9 minutes ago	Exited (0) 5 minutes ago		cc
8ae246c59541	ubuntu	"bash"	24 hours ago	Up 13 minutes		cv1
99bac679a6c8	mysql	"docker-entrypoint.s..."	24 hours ago	Exited (1) 24 hours ago		vigilant_
vaughan						
f414f4e350bb	nginx	"/docker-entrypoint..."	24 hours ago	Exited (0) 24 hours ago		condescen
ding_moore						
ff0dcb479cbf	mysql	"docker-entrypoint.s..."	2 days ago	Exited (255) 24 hours ago	33060/tcp, 0.0.0.0:3307->3306/tcp	mysql-tes
t						
e7fccf0dcc06	nginx	"/docker-entrypoint..."	2 days ago	Exited (0) 2 days ago		exciting_
haibt						
b9da867a159b	nginx	"/docker-entrypoint..."	2 days ago	Exited (0) 2 days ago		beautiful
_jennings						
2f7c1f4cb3e4	nginx	"/docker-entrypoint..."	2 days ago	Exited (0) 2 days ago		condescen
ding_hertz						
92c56770ba54	nginx	"/docker-entrypoint..."	2 days ago	Exited (0) 2 days ago		infallibl
e_tharp						
5813b30215fa	mongo	"docker-entrypoint.s..."	2 days ago	Exited (0) 2 days ago		my_app
66623d46ac69	mongo	"docker-entrypoint.s..."	6 days ago	Exited (255) 2 days ago	27017/tcp, 0.0.0.0:80->8082/tcp	DB-app
839710af1c55	ubuntu	"bash"	6 days ago	Exited (0) 6 days ago		adoring_r

RAM 0.82 GB CPU 0.08% Disk: 4.49 GB used (limit 1006.85 GB)

```
Terminal

bin boot data dev etc home lib lib64 media mnt opt proc root run/sbin/srv/sys tmp usr/var
root@8ae246c59541:/# cd data
root@8ae246c59541:/data# cat hostfile.txt
Hello
from
HOST
machine
root@8ae246c59541:/data# echo "Modified inside container" >> /data/hostfile.txt
root@8ae246c59541:/data# exit
exit
PS C:\Users\VICTUS> -
```

Step 1: Create a Directory on Host Machine

Step 2: Create a File on Host System

Step 3: Run Ubuntu Container with Volume Mount

Step 4: Access Mounted Directory Inside Container

Step 5: Modify the File Inside Container

Step 6: Exit the Container

Restarting the same container and make modification

```
iskov
PS C:\docker-volume-demo> docker restart cc
cc
PS C:\docker-volume-demo> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
c1a698e48bee   ubuntu   "bash"    9 minutes ago   Up 7 seconds   cc
8ae246c59541   ubuntu   "bash"    24 hours ago   Up 13 minutes   cv1
PS C:\docker-volume-demo> cd hostfile.txt
cd : Cannot find path 'hostfile.txt' because it does not exist.
At line:1 char:1
+ cd hostfile.txt
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (hostfile.txt:String) [Set-Location], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\docker-volume-demo> docker run -it --name cc1 -v C:\docker-volume-demo:/data ubuntu bash
root@dc4b1444d11e:/# ls
bin boot data dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@dc4b1444d11e:/# cd data
root@dc4b1444d11e:/data# cat hostfile.txt
Hello
from
HOST
RAM 0.84 GB CPU 0.08% Disk: 4.49 GB used (limit 1006.85 GB)
```

Checking the same file in different terminal

```
updating the file here
root@dc4b1444d11e:/data# echo "modified it again " >> /data/hostfile.txt
root@dc4b1444d11e:/data# cat hostfile.txt
Hello
from
HOST
machine
updating the file here
modified it again
root@dc4b1444d11e:/data# exit
exit
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-Content], ParameterBindingException
+ FullyQualifiedErrorId : NamedParameterNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\docker-volume-demo> type C:\docker-volume-demo\hostfile.txt
Hello
from
HOST
machine
灵慈棉市璵故映日 故敲洛摯晚敦 瑩愠慧潼否
PS C:\docker-volume-demo> 
RAM 0.83 GB CPU 0.17% Disk: 4.49 GB used (limit 1006.85 GB)
```